Question	Answer	Marks	Partial Marks
15(a)	6.6×10^4	**1**	
15(b)	4.07×10^8	**2**	**M1** for figs 407 or for 0.37×10^8 or 37×10^7 or $10^7(3.7 \times 10 + 3.7)$ or $10^8(3.7 + 3.7 \div 10)$ oe
16	$x \leqslant 2.5$ $y > 3$ $y \leqslant 4$ $y \leqslant 2x$	**4**	**B3** for answer $x < 2.5$ $y \geqslant 3$ $y < 4$ $y < 2x$ OR **B1** for each If 0 or 1 scored, instead award **SC2** for recognition of $x = 2.5$, $y = 3$, $y = 4$, $y = 2x$ If 0 scored, **SC1** for recognition of $y = 2x$
17(a)	91	**2**	**M1** for $7(3^2 + 2^2)$ oe
17(b)	$[\pm]\sqrt{\frac{I}{M} - c^2}$ or $[\pm]\sqrt{\frac{I - Mc^2}{M}}$ final answer	**3**	**M1** for correctly dividing by M **M1** for correctly isolating term in k^2 **M1** for correctly taking square root Maximum **M2** if answer is incorrect
18	[a =] 4 [b =] 16 [c =] 10	**4**	**B3** for two correct nfww OR **B2** for $4x^2 + 20x + 25$ or **B1** for three terms correct from $4x^2 + 10x + 10x + 25$ **M1** for $2(2x + 5) + 5$ soi
19	$-\frac{8}{3}$ or $-2\frac{2}{3}$ oe	**3**	**M2** for $-x = 2(x + 4)$ oe or **M1** for $(3^{-1})^x$, 3^{-x}, $(3^2)^{x+4}$, $3^{2(x+4)}$ or $\left(\frac{1}{3}\right)^{-2(x+4)}$ oe